Movement Across Cell Membranes (Chapter 7)

7 Water diffuses into and out of the cell directly through the plasma membrane and by facilitated diffusion through aquaporins. (See Figure 7.10.)

8 By passive transport, the CO_2 used in photosynthesis diffuses into the cell, and the O_2 formed as a by-product of photosynthesis diffuses out of the cell. Both solutes move down their concentration gradients. (See Figures 7.11 and 10.21.)

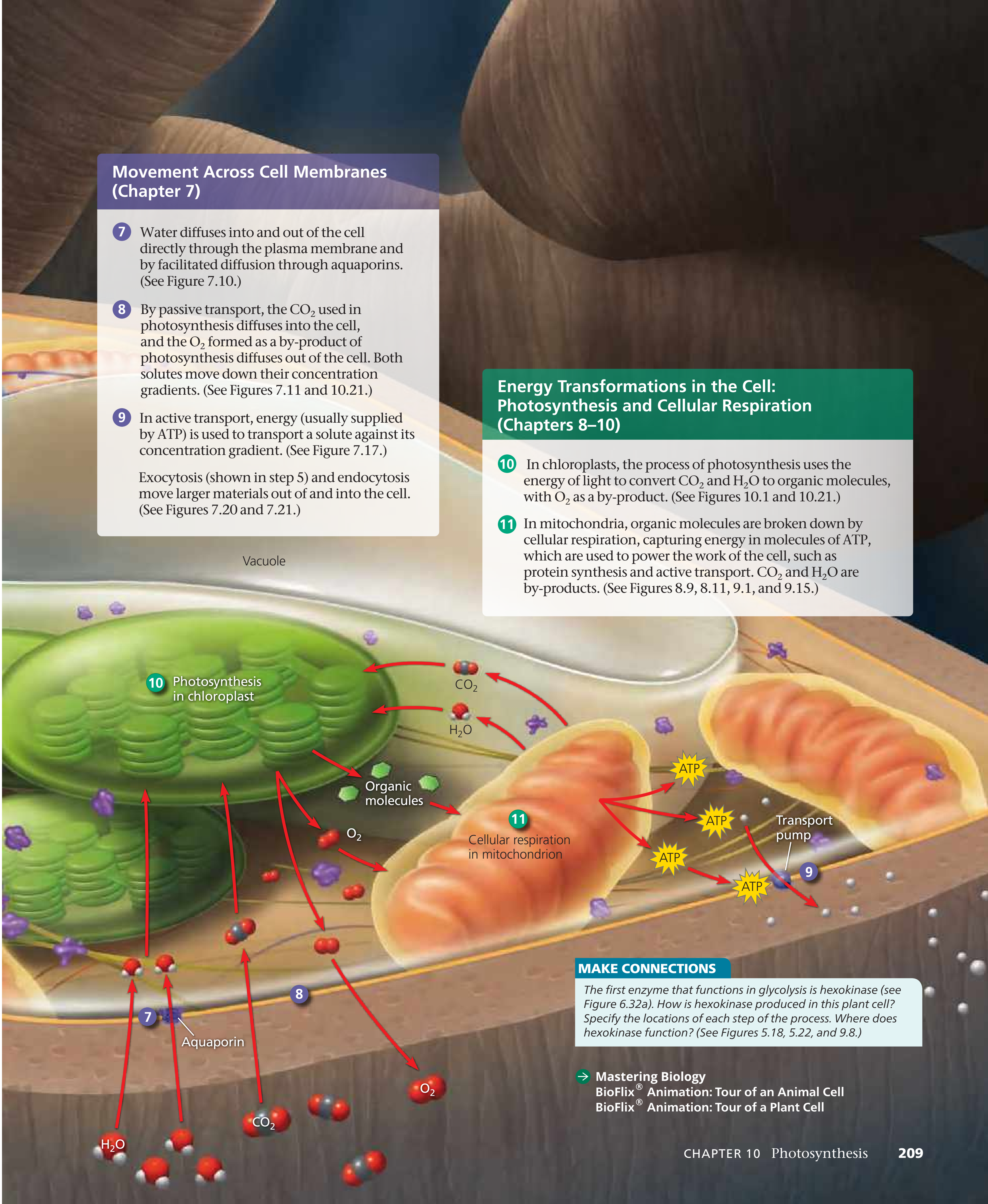
9 In active transport, energy (usually supplied by ATP) is used to transport a solute against its concentration gradient. (See Figure 7.17.)

Exocytosis (shown in step 5) and endocytosis move larger materials out of and into the cell. (See Figures 7.20 and 7.21.)

Energy Transformations in the Cell: Photosynthesis and Cellular Respiration (Chapters 8–10)

10 In chloroplasts, the process of photosynthesis uses the energy of light to convert CO_2 and H_2O to organic molecules, with O_2 as a by-product. (See Figures 10.1 and 10.21.)

11 In mitochondria, organic molecules are broken down by cellular respiration, capturing energy in molecules of ATP, which are used to power the work of the cell, such as protein synthesis and active transport. CO_2 and H_2O are by-products. (See Figures 8.9, 8.11, 9.1, and 9.15.)



MAKE CONNECTIONS

The first enzyme that functions in glycolysis is hexokinase (see Figure 6.32a). How is hexokinase produced in this plant cell? Specify the locations of each step of the process. Where does hexokinase function? (See Figures 5.18, 5.22, and 9.8.)

Mastering Biology
BioFlix® Animation: Tour of an Animal Cell
BioFlix® Animation: Tour of a Plant Cell